

Muhammad Umar

FPGA Design Engineer

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Professional Summary

Computer Engineering graduate student and FPGA Design Engineer with experience in RTL design, SystemVerilog/UVM verification, and FPGA-based deep learning. Skilled in Verilog/SystemVerilog, Vivado, Vitis HLS, PYNQ/Zynq platforms, SystemRDL, and deploying DNN models for real-time computer-vision applications.

Technical Skills

- **FPGA/ASIC/SoC Life Cycle:** Architecture, design and development, simulation, integration, and testing.
- **Programming and HDLs:** Python, C, Verilog, SystemVerilog, SystemRDL, MATLAB, Git, Bitbucket.
- **Developer Tools:** AMD Vivado Design Suite, Vitis HLS, ISE, ModelSim/QuestaSim, Synopsys VCS, MATLAB, Visual Studio Code, Jupyter Notebook, Google Colab, GVim, Android Studio.
- **FPGA Boards:** Basys 3, Nexys-7, Zynq-7000, PYNQ-Z1.
- **Design Skills:** RTL coding, simulation, place and route, debugging, testing, verification, APB, SPI, I2C, UART, basic Ethernet, and UVM.

Professional Experience

Design Verification Engineer – CoMira Solutions Inc.

Dec 2023 – Dec 2024

- Completed six months of structured training covering combinational, sequential, and finite-state-machine logic design.
- Contributed to the verification of digital designs using the UVM methodology.
- Studied the fundamentals of MAC and PCS modules based on the IEEE 802.3 specification.
- Explored SystemRDL for automating design-verification flows.

FPGA Intern – Neurocomputation Lab, National Center of Artificial Intelligence

Nov 2022 – Aug 2023

- Applied classical computer-vision techniques for edge detection and developed RTL in Verilog.
- Deployed DNNs on PYNQ and prototyped FPGA inference flows using MATLAB.

University Experience

Instructional Student Assistant – ECE 427/L: Digital Design with Verilog and SystemVerilog

Aug 2026 – Present

- Assisted with theory and laboratory instruction in digital design, Verilog, SystemVerilog, RTL design, simulation, and FPGA implementation.
- Supported students with debugging, waveform analysis, design verification, digital-logic development, and laboratory assignment evaluation.

IT Student Assistant – Mike Curb College of Arts, Media, and Communication

Sep 2026 – Present

- Provide technical support for faculty, staff, and students by assisting with hardware, software, and classroom-technology issues.
- Troubleshoot user problems, document resolutions, and support day-to-day IT operations.

Instructional Student Assistant – ECE 220/L: Theory of Digital Systems

Feb 2026 – May 2026

- Supported instruction and laboratory activities covering digital logic, combinational and sequential circuits, and Verilog fundamentals.
- Assisted students with laboratory exercises, simulation, debugging, and digital-design concepts.

Education

California State University, Northridge

Northridge, CA / Jan 2025 – Present

Master of Science in Computer Engineering; CGPA: 3.7/4.0

Relevant Coursework: Digital Design with Verilog and SystemVerilog, System-on-Chip (SoC), Computer Architecture, Digital Electronics, Image Processing, Computer Arithmetic and Design, Advanced Switching Theory.

NED University of Engineering and Technology

Oct 2019 – Aug 2023

Bachelor's Thesis – Design of Structural Health Monitoring System on FPGA

- Implemented FPGA-based edge detection using Verilog for structural-crack analysis.
- Trained and deployed a YOLOv7 model on PYNQ-Z1 for real-time crack detection.

Academic Projects

FIFO and ALU Verification Environment

- Designed an ALU and synchronous FIFO in SystemVerilog and integrated them through a wrapper.
- Verified the design using a SystemVerilog UVM environment with functional coverage, code coverage, constrained-random testing, and corner-case verification.

AMBA APB Register Subsystem Verification

- Designed a configurable APB-based register subsystem in SystemVerilog and verified it using a UVM RAL environment.
- Used SystemRDL to define the register map and automate register model/documentation generation.

Traffic Light Controller

- Designed a parameterized FSM-based traffic-light controller in SystemVerilog for main-highway, sub-highway, and train-crossing scenarios.
- Implemented sensor-driven signal control, main-highway priority, train override logic, and configurable light timing.

Elevator System on Nexys-7 Board

- Implemented a Verilog-based elevator controller and simulated the design in Vivado.
- Prepared the design for FPGA deployment on the Nexys-7 board.

Vending Machine FSM

- Designed a finite-state-machine controller in Verilog to model item selection, input states, and transaction flow.
- Simulated and verified the FSM behavior in Vivado.